Minimalist Foundations

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https://kxhoge.github.io/resources.html

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Course outline

Monday	From X-bar theory to Bare Phrase Structure: The emergence of Merge
Tuesday	Merge, Move, and Agree
Wednesday	The expansion of Merge: External / Internal Merge and Phase theory
Thursday	The Strong Minimalist Thesis, Merge, and the problem of labelling
Friday	The Miracle Creed: Can syntax be reduced to Merge?

The expansion of Merge: External / Internal Merge and Phase theory

Outline

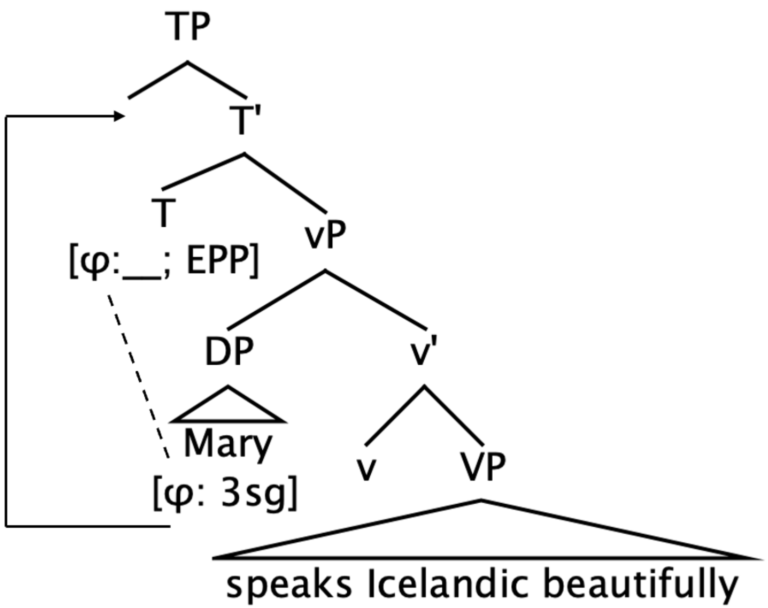
1. More on Agree
2. Displacement – from imperfection to virtual conceptional necessity
3. The need for phases
4. Summary and outlook

The expansion of Merge: External / Internal Merge and Phase theory

More on Agree

(1) Mary speaks Icelandic beautifully.

(2)



The expansion of Merge: External / Internal Merge and Phase theory

More on Agree

- *Try your hand at some syntactic analysis*

- Consider the Icelandic example in (3):

(3) Henni leidd**ust** strák**arnir**.

her(D) bored(3pl) the boys(Npl)

`She found the boys boring.'

Henni 'her' and *strákarnir* 'the boys' are the external and internal arguments, respectively, of *leiddust* 'bored'. Assume that *henni* is merged in Spec*v*P and *strákarnir* in the complement position of V. Draw the syntactic tree structure for the sentence, indicating the distribution of the features relevant to Agree and Move.

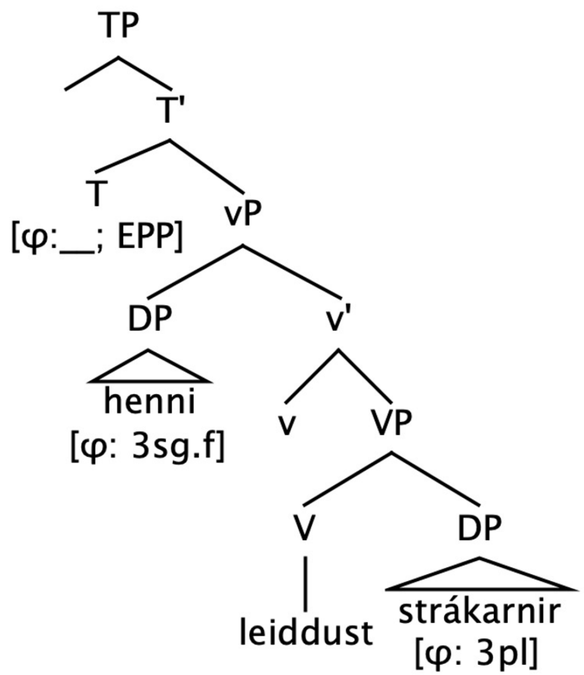
What problems, if any, do you encounter?

The expansion of Merge: External / Internal Merge and Phase theory

More on Agree

- *Try your hand at some syntactic analysis*

(4)



Problem 1: The structurally closest DP (the dative experiencer) does not value the phi-features of T; instead, agreement is established with the lower nominative DP.

Problem 2: The DP that values T's phi-features is not the one that moves to SpecTP to satisfy the EPP, i.e. subject movement appears not to be contingent on Agree.

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The expansion of Merge: External / Internal Merge and Phase theory

Displacement – from imperfection to virtual conceptual necessity

- In the Minimalist tradition, **displacement** (also called dislocation) refers to the phenomenon whereby a syntactic element is interpreted in one position but pronounced in another as the result of movement.

(5) a. They elected an unpopular candidate.
  b. An unpopular candidate was elected.

- If FL is organised as a maximally economical system that generates linguistic expressions legible at the sensorimotor and conceptual-intentional interfaces, C_{HL} should do no more than is necessary to satisfy the legibility conditions imposed by PF and LF. Displacement therefore appears to be an imperfection, introducing redundancy and additional computational complexity beyond what is required by the interfaces.

The expansion of Merge: External / Internal Merge and Phase theory

Displacement – from imperfection to virtual conceptual necessity

The dislocation property (34b) is another apparent imperfection. In (35), for example, the phrase *an unpopular candidate* is in the natural position to be interpreted as object of *elect* in (35a,b) but not in (35c,d), though the interpretation is in relevant respects the same.

(35) a. they [elected an unpopular candidate]
b. there was [elected an unpopular candidate]
c. an unpopular candidate was elected
d. there was an unpopular candidate elected

(Chomsky 2000: 119)

The expansion of Merge: External / Internal Merge and Phase theory

Displacement – from imperfection to virtual conceptual necessity

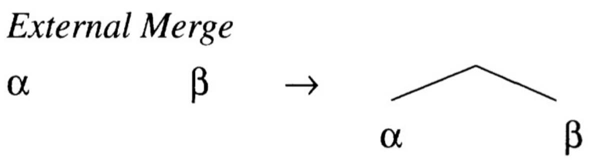
- While Chomsky (2001: 23) continues to analyse Move as a compound operation comprising Agree, pied-piping and Merge ('The compound operation Move consists of Agree, pied-piping, and Merge'), **Chomsky (2004**: 110) **abandons Move as a syntactic operation**, arguing instead that 'Merge is unconstrained, therefore either external or internal'.

The expansion of Merge: External / Internal Merge and Phase theory

Displacement – from imperfection to virtual conceptual necessity

- **External Merge** combines two distinct syntactic objects to create a new syntactic object.

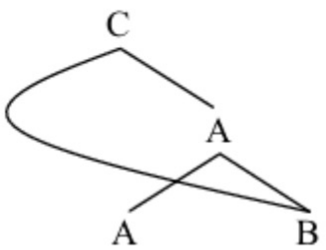
(6)



(Citko 2005: 475, ex. (1))

- **Internal Merge** combines a syntactic object with a subpart of that syntactic object, remerging the subpart at the root of the structure that already contains it.

(7)



(Citko & Gračanin-Yüksek 2020: 2, ex. (1b))

The expansion of Merge: External / Internal Merge and Phase theory

Displacement – from imperfection to virtual conceptual necessity

- ‘Accordingly, displacement is not an “imperfection” of language; its absence would be an imperfection’ (Chomsky 2004: 110).

- ‘…there are two subcases of the operation Merge. Given A, we can merge B to it from outside A or from within A; these are external and internal Merge, the latter the operation called “Move”, which therefore also “comes free”, yielding the familiar displacement property of language. That property had long been regarded, by me in particular, as an “imperfection” of language that has to be somehow explained, but in fact it is a virtual conceptual necessity’ (Chomsky 2005: 12).

The expansion of Merge: External / Internal Merge and Phase theory

The need for phases

- Reconceptualising Move as Internal Merge raises the question of whether, and in what way, Merge must be constrained. If External and Internal Merge are instances of the same basic operation, what accounts for the apparent asymmetry between them, as reflected in Chomsky's (1995: 216) statement that 'Merge is costless for principled reasons, movement is not'?

The expansion of Merge: External / Internal Merge and Phase theory

The need for phases

- The costlessness of Merge follows from its necessity for a derivation to converge. Since insufficient application of Merge results in no derivation at all, Merge is not an operation to which considerations of economy apply.

  'A derivation converges only if [Merge] has applied often enough to leave us with just a single object, also exhausting the initial numeration …
  If Select [the procedure that selects a lexical item from the numeration, reduces its index by one and introduces it into the derivation] does not exhaust the numeration, no derivation is generated … Insufficient application of Merge has the same property, since the derivation then fails to yield an LF representation at all; again, no derivation is generated …
  The operations Select and Merge are "costless"; they do not fall within the domain of discussion of convergence and economy.(Chomsky 1995: 207-208).

The expansion of Merge: External / Internal Merge and Phase theory

The need for phases

- The costlessness of Merge, together with the reconceptualisation of Move as Merge + Agree in Chomsky (2000), motivates the principle that Merge / Agree pre-empts Move (cf. Chomsky 2000: 104).

The expansion of Merge: External / Internal Merge and Phase theory

The need for phases

- Empirical evidence for **Merge trumping Move** is supplied by examples like (8):

(8) a. There seems [$_{TP}$ ~~there~~ to be someone in the room].
(Chomsky 1995: 317, ex. (170))

b. *There seems [$_{TP}$ someone to be ~~someone~~ in the room].
(Chomsky 1995: 317, ex. (169))

(9) N = {(there, 1), (seems, 1), (to, 1), (be, 1), (someone, 1), (in, 1), (the, 1), (room, 1), (T, 2)}

- Both derivations involve a stage at which embedded T requires a nominal in its specifier to satisfy the EPP requirement.

(10) [$_{TP}$ to be [someone in the room]]

At that stage, two possibilities arise: (i) Merge *there* OR (ii) Move *someone*. The ungrammaticality of (8b) suggests that at any given stage of the derivation, the 'best possible (most economical) move available' at that stage is chosen (Chomsky 1995: 319), favouring Merge over Move.

The expansion of Merge: External / Internal Merge and Phase theory

The need for phases

- But if Move is costlier than Merge, what explains the acceptability of (11a) (alongside (11b)? Why does the Merge of *there* not pre-empt the Move of *someone*?

(11) a. There is a possibility that [$_{TP}$ someone is ~~someone~~ in the room].
b. The possibility is that [$_{TP}$ there is someone in the room]

- Chomsky's (2000) solution relies on the idea that at the point of the derivation when a nominal needs to be merged into the specifier of the embedded TP, expletive *there* is not available since it is not part of the lexical (sub-)array to which the derivation has access.

The expansion of Merge: External / Internal Merge and Phase theory

The need for phases

(11) a. There is a possibility that [$_{TP}$ someone is ~~someone~~ in the room].

b. The possibility is that [$_{TP}$ there is someone in the room]

- 'Suppose we select [a lexical array; LA] as before ... Suppose further that at each stage of the derivation a subset LA_i is extracted, placed in active memory ('the workspace'), and submitted to the procedure L [= narrow syntax]. When LA_i is exhausted, the computation may proceed if possible. Or it may return to LA and extract LA_j, proceeding as before. The process continues until it terminates. Operative complexity in some natural sense is reduced, with each stage of the derivation accessing only part of LA. If the subarray in active memory does not contain Expl, then Move can take place in the corresponding stage; if it does, Merge of Expl preempts Move' (Chomsky 2000: 106).

The expansion of Merge: External / Internal Merge and Phase theory

The need for phases

- Chomsky (2000) thus replaces numeration with **lexical array**. Like a numeration, a lexical array is a collection of lexical items chosen for a derivation, but, unlike a numeration, a lexical array lacks indices that indicate how many times a given lexical item is to be selected.
- A lexical array can be divided into sub-arrays, associated with syntactic **phases**.

> LA_i should determine a natural syntactic object SO, an object that is relatively independent in terms of interface properties. On the "meaning side", perhaps the simplest and most principled choice is to take SO to be the closest syntactic counterpart to a proposition: either a verb phrase in which all theta-roles are assigned or a full clause including tense and force. Call these objects propositional …
>
> LA can then be selected straightforwardly: LA_i contains an occurrence of C or of v, determining clause or verb phrase … Take a **phase** of a derivation to be an SO derived in this way by choice of LA_i. **A phase is CP or vP**' (Chomsky 2000: 106; emphasis added).

The expansion of Merge: External / Internal Merge and Phase theory

The need for phases

- Phases define **cyclic domains of syntactic computation**. At the end of each phase, the syntactic object constructed thus far is transferred to the sensorimotor (SM) and conceptual-intentional (C-I) interfaces.

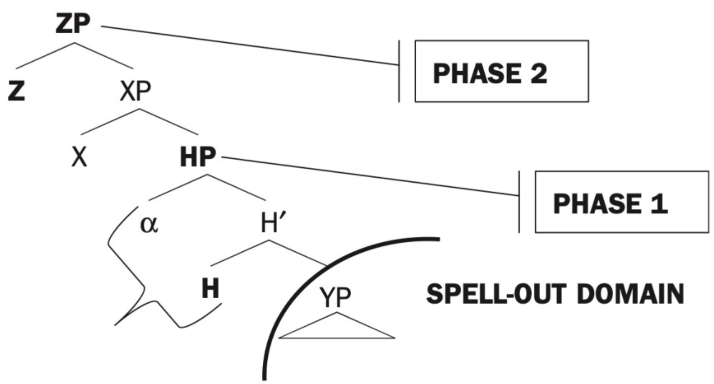
- Transfer renders a portion of the syntactic derivation inaccessible to subsequent syntactic computation. Phases therefore impose a cyclic organisation on the derivation, partitioning the syntactic computation into successive domains rather than allowing the entire structure to be built and interpreted at once.

The expansion of Merge: External / Internal Merge and Phase theory

The need for phases

- Syntactic locality can now be expressed in phase-theoretic terms through the Phase Impenetrability Condition (PIC), which constrains access to material contained within a completed phase.
- **Phase Impenetrability Condition**
  'In phase α with head H, the domain of H is not accessible to operations at ZP; only H and its edge are accessible to such operations' (Chomsky 2001: 13-14).

(12)



(Citko 2014: 34, ex. (26b))

The expansion of Merge: External / Internal Merge and Phase theory

The need for phases

- One consequence of the PIC is that a *wh*-phrase displaced to clause-initial position must proceed via the edge of an intervening *v*P phase; that is, it must undergo Internal Merge in Spec*v*P before undergoing further Internal Merge to SpecCP, thereby avoiding inaccessibility under the PIC.

The expansion of Merge: External / Internal Merge and Phase theory

The need for phases

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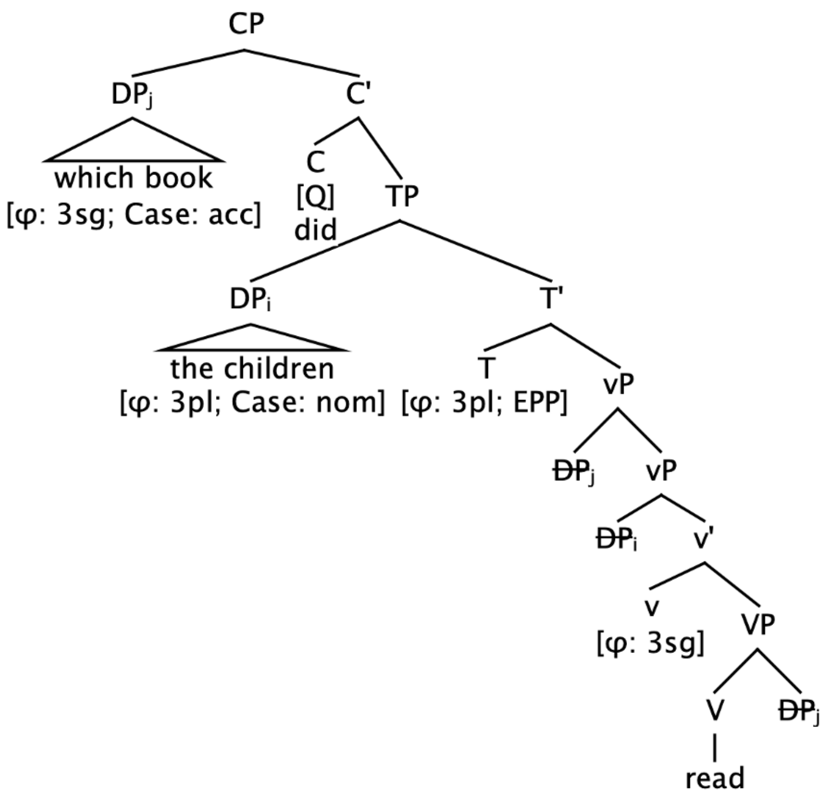

The expansion of Merge: External / Internal Merge and Phase theory

The need for phases

(14) a. Which book did the children read?

b. [$_{CP}$ Which book C$_{[Q]}$ [$_{TP}$ the children [$_{v*P}$ <which book> [<the children> v* [$_{VP}$ read <which book>]]]]]

The expansion of Merge: External / Internal Merge and Phase theory

Summary and outlook

- With the recognition of External and Internal Merge (Chomsky 2004), syntactic movement in effect comes for free, leading to a new understanding of displacement as a virtual conceptual necessity rather than an imperfection.
- The reconceptualisation of Move as Internal Merge and the emergence of phase theory may be seen as developments that go hand in hand: while movement is reduced to the independently motivated operation Merge, phases provide the cyclic architecture that constrains Internal Merge and derives syntactic locality through the Phase Impenetrability Condition.

- With Phase Theory requiring a displaced XP to pass through the edge of each intervening phase in accordance with the PIC, the question arises as to whether the satisfaction of formal features is the only factor that drives Internal Merge.

The expansion of Merge: External / Internal Merge and Phase theory

References

Chomsky, Noam. 1995. *The Minimalist program*. Cambridge, MA: MIT Press.

Chomsky, Noam. 2000. Minimalist inquiries: The framework. In Roger Martin, David Michaels & Juan Uriagereka (eds.), *Step by step: Essays on Minimalist syntax in honor of Howard Lasnik*, 89–155. Cambridge, MA: MIT Press.

Chomsky, Noam. 2001. Derivation by Phase. In Michael Kenstowicz (ed.), *Ken Hale: A life in language*, 1–52. Cambridge, MA: MIT Press.

Chomsky, Noam. 2004. Beyond explanatory adequacy. In Adriana Belletti (ed.), *Structures and beyond: The cartography of syntactic structures*, vol. 3, 104–131. New York & Oxford: Oxford University Press.

Chomsky, Noam. 2005. Three factors in language design. *Linguistic Inquiry* 36(1), 1–22. doi:10.1162/0024389052993655. https://direct.mit.edu/ling/article/36/1/1-22/250.

Citko, Barbara. 2005. On the nature of Merge: External Merge, Internal Merge, and Parallel Merge. *Linguistic Inquiry* 36(4), 475–496. doi:10.1162/002438905774464331. https://direct.mit.edu/ling/article/36/4/475-496/272.

Citko, Barbara. 2014. *Phase theory: An introduction*. Cambridge: Cambridge University Press.

Citko, Barbara & Martina Gračanin-Yuksek. 2020. *Merge: Binarity in (multidominant) syntax*. Cambridge, MA: MIT Press. doi:10.7551/mitpress/12800.001.0001. https://direct.mit.edu/books/book/5006/MergeBinarity-in-Multidominant-Syntax.